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Sheet 1 أحمد عماد محمود زايد (5L-T3)

1.1 Define the following items:

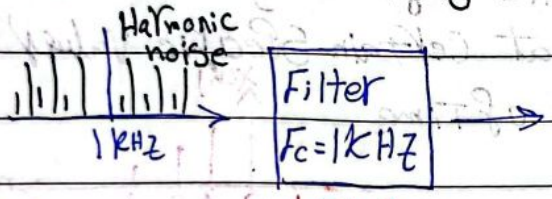
- **Signal** → Physical quantity versus time or space or any independent variable or variables.

$S1 = 5t$ $S2 = 20t^2$ $S(x,y) = 20x^2 + 2xy$

(x,y) → independent variables

$S(x,y)$ → dependent variables

- **System** → Physical device that performs operation on signal



1.2

↳ Reduce noise of signal

- **Signal Processing** → operation performs on signal

There are 2 Type:

1. Analog Signal Processing (ASP)



2. Digital Signal Processing (DSP)



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1.3 Classify different Signal Types

* multi Channel & multi dimensional

Signal generate From multi Source (Sensor)

$$X(t)=\begin{bmatrix} x_1(t) & \text{current} \\ x_2(t) & \text{Temp} \\ x_3(t) & \text{noise} \end{bmatrix}$$

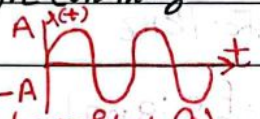
Signal is Function of Signal independent variables

$X(t)=3t$

$X(v,z)=2v^2+3z$

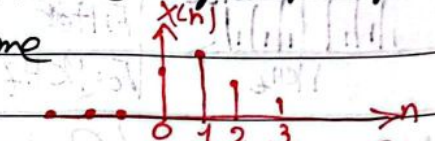
* Continuay Time & discrete Time

Analog Signal $X_a(t)$ Every value of Time & Take Value in continuay interval



$x(t)=A\sin(2\pi ft+\theta)$

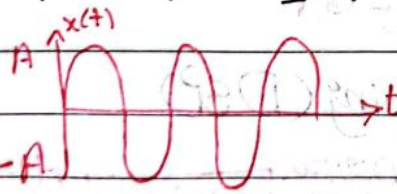
Are define only at certain Specific values of Time



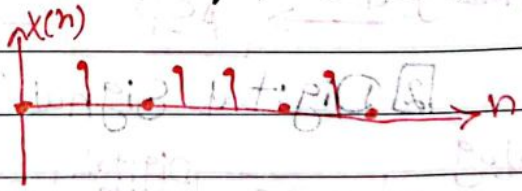
$X(n)=A\sin(2\pi Fn+\theta)$

* Continuay Valued & discrete value

Take All Possible Values Finite or infinite

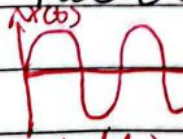


Take one Value From Finite of Possible values



* deterministic & Random

Any Signal describe By mathematical Expression, Table of data & graph



$X_a(t)=A\sin(2\pi ft+\theta)$

Signal Can't describe By Mathematical Expression Table of data & graph

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1.5 State and Explain the Sampling theorm

↳ For Signal with max Freq. (F_{max}) Can Sampled Signal with Sample Rate (F_s) Must greater than Twice max Freq.

$F_s \geq 2F_{max}$ (without Aliasing)

Nyquist Rate ⇒ Min Sampling Rate For Sample Signal without Aliasing.

where $F_s = 2F_{max}$

1.6 Determine which the following Sinusoidal are Periodic and Compute the Fundamental Period.

ⓐ $x(t) = 3\cos(5t + \frac{\pi}{6})$

$x(t) = A\cos(2\pi ft + \theta)$

$A=3$, $2\pi f = 5 \rightarrow f = \frac{5}{2\pi}$ Hz, $T_p = \frac{2\pi}{5}$ Sec

$x(t+T_p) = 3\cos(5(t+T_p) + \frac{\pi}{6})$

$= 3\cos(5t + 5T_p + \frac{\pi}{6})$

$= 3\cos(5t + \frac{\pi}{6}) = x(t)$

Periodic Signal every $\frac{2\pi}{6}$ Sec.

~~$x(n) = 3\cos(5nT_s + \frac{\pi}{6})$~~

~~$x(n) = A\cos(2\pi Fn + \theta)$~~

~~$2\pi F = 5T_s \rightarrow F = \frac{5T_s}{2\pi}$~~

Fundamental Period

$T_p = \frac{2\pi}{5}$ Sec.

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(b) $X(t) = 3\cos(100\pi t)$

$x(t) = A\cos(2\pi f t + \theta)$

$f = 50\,Hz \rightarrow T_p = 20\,mSec$

$x(t+T_p) = 3\cos(100\pi t + \frac{100\pi}{50}) = 3\cos(100\pi t)$

∴ Signal is Periodic every 20mSec.

$X(t) \rightarrow X(n)$

$t \rightarrow nTs$ $\quad F_s \geq 2F_{max}$

$F_{smin} = 2 \times 50 = 100\,Hz$

$X(n) = 3\cos(100\pi n T_s)$

$= 3\cos(\frac{100\pi n}{100}) = 3\cos(n\pi)$

$F = \frac{f}{f_s} = \frac{50}{100} = \frac{1}{2} = \frac{k}{N_p} \rightarrow N_p = 2\,Samples$ *

(c) $x(n) = \cos(\frac{30}{105}\pi n)$

$x(n) = A\cos(2\pi F n + \theta)$

$2F = \frac{30}{105} \rightarrow F = \frac{30}{210}$ cycle/Sample

$F = \frac{1}{7} = \frac{k}{N_p} \rightarrow N_p = 7\,Samples$

$x(n+N_p) = \cos(\frac{30}{105}\pi(n+N_p))$

$= \cos(\frac{30}{105}n\pi + \frac{30}{105} \times 7\pi)$

$= \cos(\frac{30}{105}\pi n) = x(n)$

Signal is Periodic every 7 Samples.

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1.7 Consider the following analog Sinusoidal

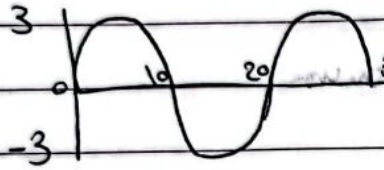
$X_a(t) = 3\sin(100\pi t)$

ⓐ Sketch the signal for [0, 30ms]

$X(t) = 3\sin(100\pi t)$

$x(t) = A\sin(2\pi f t + \theta)$

$A=3,\ F=50\,Hz,\ T_p=20m\,Sec,\ \theta=0^\circ$



ⓑ The signal is sampled of $F_s=300$ samples/Sec

$X(t) \longrightarrow x(n)$ $\qquad F_s \geq 2F_{max}$

$t \longrightarrow nT_s = \frac{n}{F_s}$

$x(n) = 3\sin\left(\frac{100\pi n}{300}\right) = 3\sin\left(\frac{\pi n}{3}\right)$

$F = \frac{f}{f_s} = \frac{50}{300} = \frac{1}{6} = \frac{k}{N_p} \longrightarrow N_p = 6$ samples

$T_S = \frac{1}{F_s} = \frac{1}{300}$ Sec

ⓒ Compute values in one period of $x(n)$, sketch $x(n)$ on the same diagram $X(t)$

$x(n) = 3\sin\left(\frac{\pi n}{3}\right),\ N=6 \longrightarrow n=[0,1,2,3,4,5]$

$x(0) = 0$

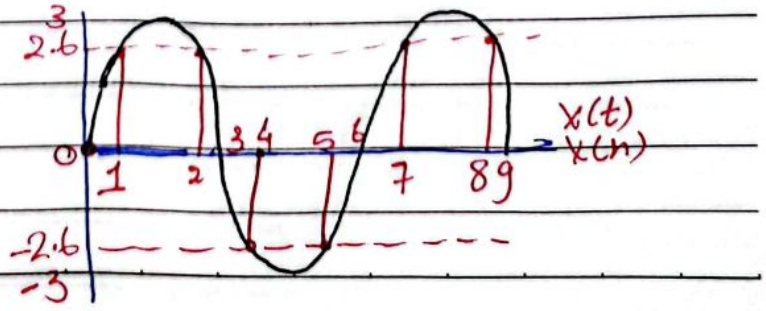
$x(1) = 2.6$

$x(2) = 2.6$

$x(3) = 0$

$x(4) = -2.6$

$x(5) = -2.6$



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① Can you find Sampling Rate Fs such that x(n) Reaches its Peak Value 3? what is Fminum for Task?

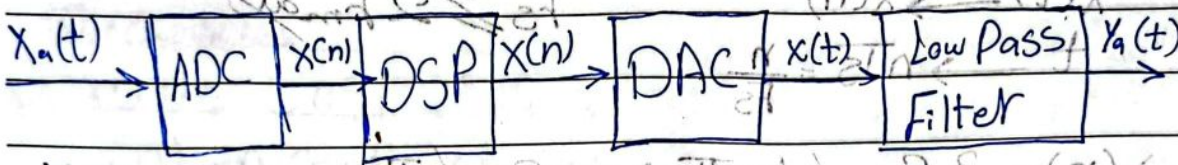
$x(n)=3$

$x(n)=3\sin\left(\frac{100\pi n}{Fs}\right)=3$

$\frac{100\pi n}{Fs}=\sin^{-1}(1)=\frac{\pi}{2}$

$\boxed{Fs=200n}$ $n=1 \rightarrow$ Fminimum

1.8 $T_1=5mS$ $T_2=1mS$ Determind the output Y_a? when input $x_a(t)=3\cos(100\pi t)+2\sin(250\pi t)$



$x_a(t)=3\cos(100\pi t)+2\sin(250\pi t)$

$nT_s=\frac{n}{Fs}=5\times10^{-3}n$

$x(n)=3\cos(100\pi*5\times10^{-3}n)+2\sin(250\pi*10^{-3}*5n)$

$=3\cos(0.5n\pi)+2\sin(1.25n\pi)$

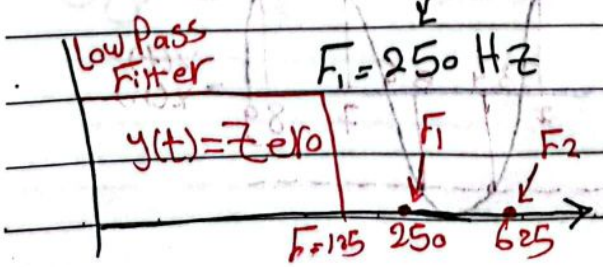
$t\rightarrow nTs \rightarrow n=\frac{t}{Ts}=t_s\cdot Fs$ $\because Ts=1msec$

$\therefore n=1000t$

$x(tFs)=3\cos(0.5\times1000\pi t)+2\sin(1.25\times1000\pi t)$

$=3\cos(500\pi t)+2\sin(1250\pi t)$

$F_1=250$ Hz $F_2=625$



Page: SHEET #2 (5L-T3) Date: (4200211)

2.2) DFT of $\{0,1,1,0\}$ and check validity of your answer By Calc. IDFT.

$$X(n)=\{0,1,1,0\}$$

$$X(k)=\sum_{n=0}^{N-1} X(n)e^{-j\frac{2\pi}{N}nk} \qquad N=4$$

$$X(k)=X(0)+X(1)e^{-j\frac{\pi}{2}k}+X(2)e^{-j\pi k}+X(3)e^{-j\frac{3\pi}{2}k}$$

(X(0) → zero, X(1) → 1, X(2) → 1, X(3) → zero)

$$X(k)=e^{-j\frac{\pi}{2}k}+e^{-j\pi k}$$

$$X(k)=\{2,-1-j,0,-1+j\}$$

AT $k=0 \rightarrow X(0)=2$

$k=1 \rightarrow X(1)=-j-1=-1-j$

$k=2 \rightarrow X(2)=-1+1=\text{zero}$

$k=3 \rightarrow X(3)=j-1=-1+j$

$$x(n)=\frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j\frac{2\pi}{N}nk}$$

$$=\frac{1}{4}\left[X(0)+X(1)e^{j\frac{\pi}{2}n}+X(2)e^{j\pi n}+X(3)e^{j\frac{3\pi}{2}n}\right] \quad (X(2)\rightarrow \text{zero})$$

$$=\frac{1}{4}\left[2+(-1-j)e^{j\frac{\pi}{2}n}+(-1+j)e^{j\frac{3\pi}{2}n}\right]$$

$$x(0)=\frac{1}{4}\left[2+(-1-j)(1)+(-1+j)(1)\right]=\text{zero}$$

$$x(1)=\frac{1}{4}\left[2+(-1-j)(j)+(-1+j)(-j)\right]$$

$$=\frac{1}{4}[2-j+1+j+1]=1$$

$$x(2)=\frac{1}{4}[2+1+j+1-j]=1$$

$$x(3)=\frac{1}{4}[2+j+1-j-1]=\text{zero}$$

is Validity

$x(n)=\{0,1,1,0\}$

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2.3) Use Result of Prob 2.2 verify Parseval's theorem of $\{0,1,1,0\} \leftarrow x(n)$

$$\sum_{n=0}^{N-1} x^2(n) = \frac{1}{N}\sum_{k=0}^{N-1} |X(k)|^2$$

$X(k) = \{2, -1-j, 0, -1+j\}$

$$L.H.S = (0)^2+(1)^2+(1)^2+(0)^2 = 2\ \#$$

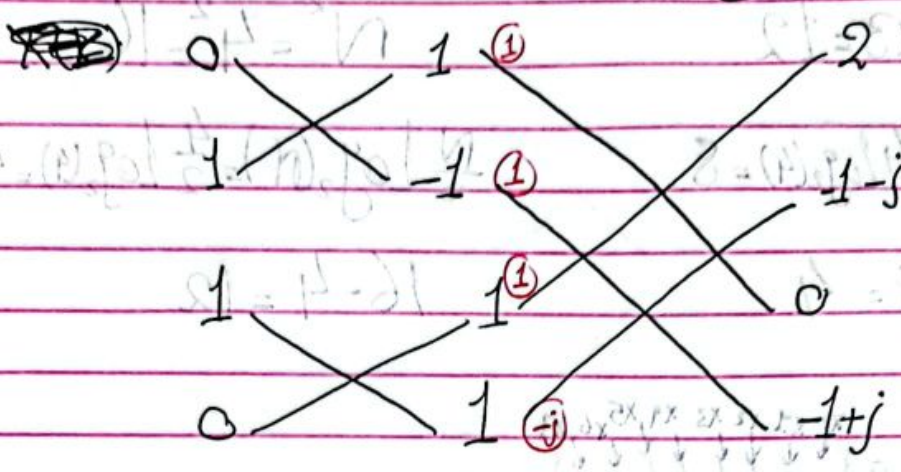
$$|X(k)| = \{2, \sqrt{2}, 0, \sqrt{2}\}$$

$$R.H.S = \frac{1}{4}\left[(2)^2+(\sqrt{2})^2+(0)^2+(\sqrt{2})^2\right] = \frac{8}{4} = 2\ \#$$

$\therefore$ Parseval's Theorem is verified.

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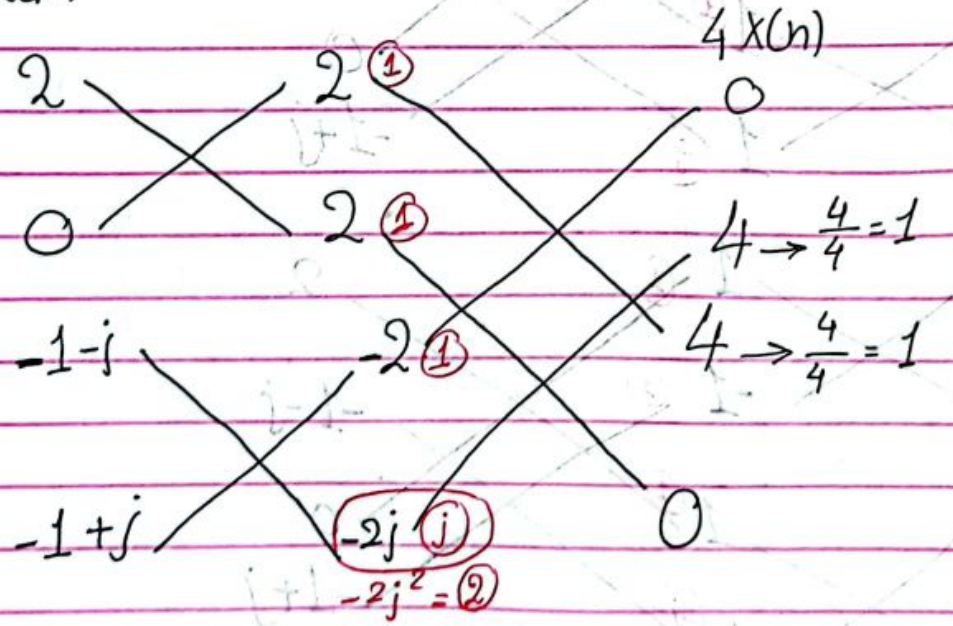
2.4) $\{0,1,1,0\} \leftarrow x(n)$ (FFT)



$x(k) = \{2, -1-j, 0, -1+j\}$

$\therefore x(k)$ in This Example $= x(k)$ in Prob(2.2)

$x(k)$

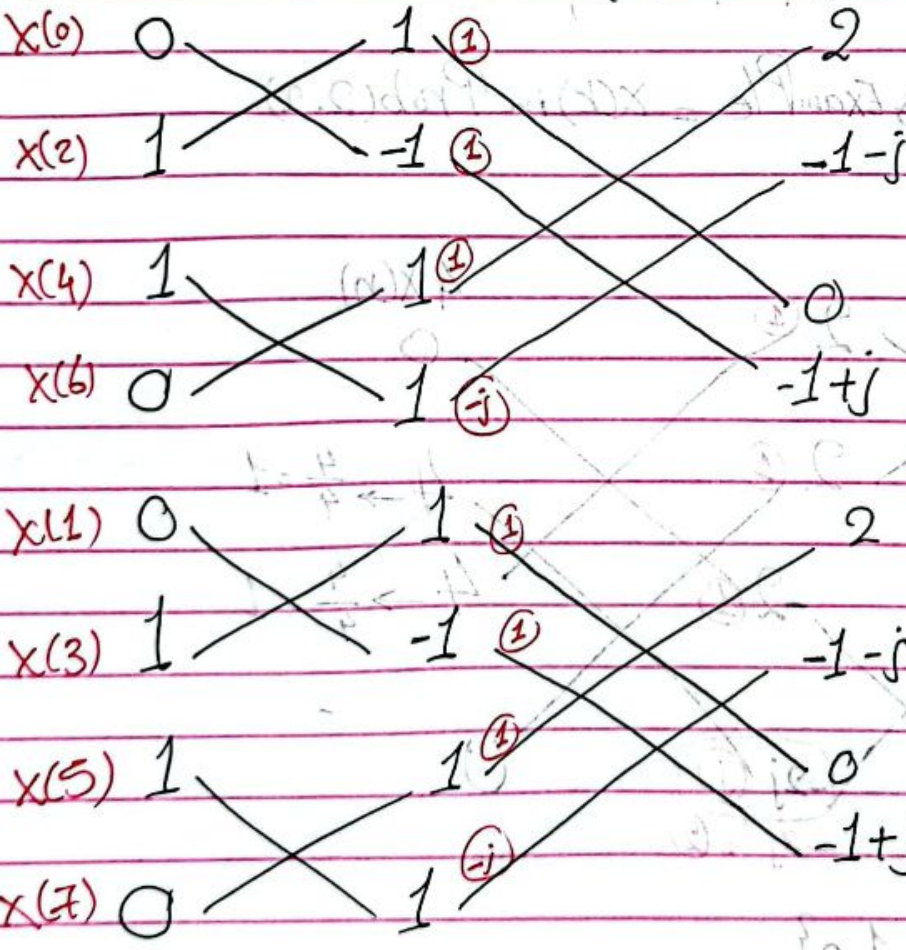


$\therefore x(n) = \{0,1,1,0\}$

Page: (N=4) عدد العمليات الحسابية المطلوبة لحساب الـ DFT و الـ FFT Date: (5/-T3)

	n.of Addition	n.of multiplication
DFT	$N(N-1)=4\times3=12$	$N^2=4^2=16$
FFT	$N\log_2(N)=4\log_2(4)=8$	$\frac{N}{2}\log_2(N)=\frac{4}{2}\log_2(4)=4$
Saving	$12-8=4$	$16-4=12$

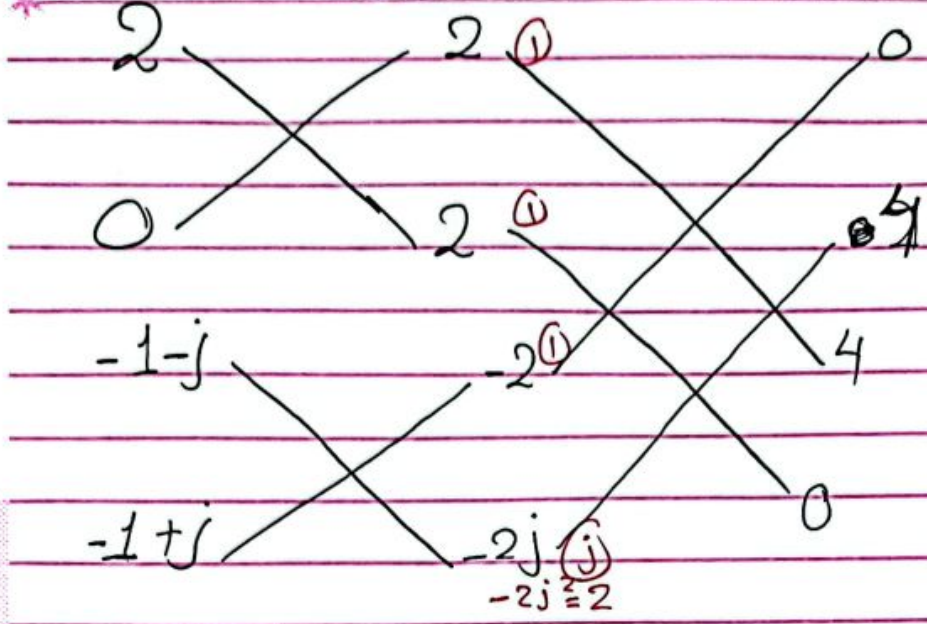
2.5) Calc. FFT of $[0,1,1,1,1,0,0]$ (x0 x1 x2 x3 x4 x5 x6 x7)

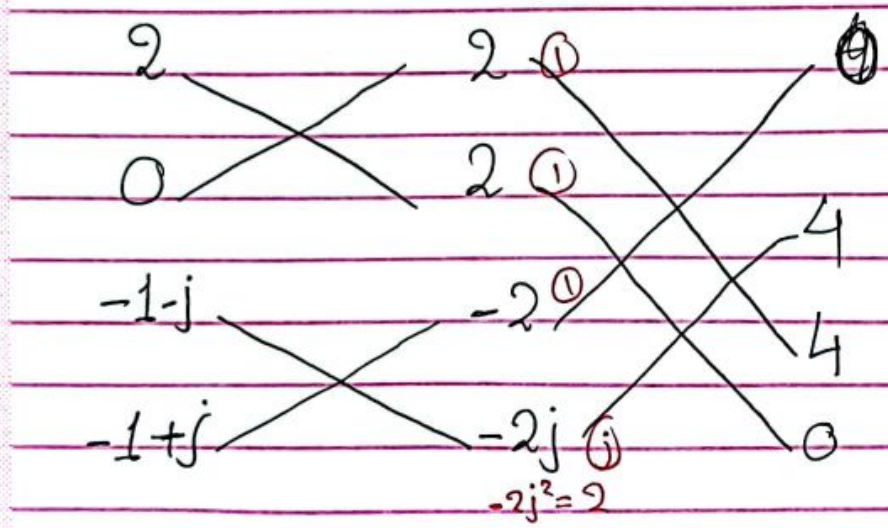


$X(K)=\{2,\ -1-j,\ 0,\ -1+j,\ 2,\ -1-j,\ 0,\ -1+j\}$

x_0 x_1 x_2 x_3 x_4 x_5 x_6 x_7

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$\therefore X(n) = \{0, 1, 1, 0, 0, 1, 1, 0\}$

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SHEET (4)

4.1: define the following Term:

* Time invariant discrete: a discrete Time Signal Said To Be invariant if it's output is independent on the Time when The input is applied $X(n) \rightarrow y(n)$ & $X(n-1) \rightarrow y(n-1)$

* Time System Linear: discrete Time ~~Signal~~ System used to Be Linear if it obayes Principles of SuperPosition

$$a_1X_1(n) \rightarrow a_1y_1(n) \quad a_2X_2(n) \rightarrow a_2y_2(n)$$

* Time System Causal Signal: System To Be Causal System if it Produces an output only where there is an input no output Before Time = Zero $h(k) = 0 \quad k < 0$

4.2: Find & Plot Z-Transform of the following discrete Time Sequence:

(a) $X(n) = \{0, 1, 3, 5, 1, 0\}$

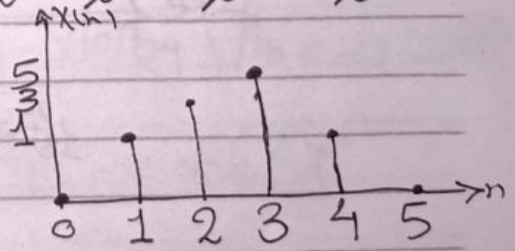
$$X(Z) = \sum_{n=-\infty}^{n-1} X(n) Z^{-n} = \sum_{n=0}^{5} X(n) Z^{-n}$$

$$= X(0)Z^{-0} + X(1)Z^{-1} + X(2)Z^{-2} + X(3)Z^{-3} + X(4)Z^{-4} + X(5)Z^{-5}$$

$$X(Z) = Z^{-1} + 3Z^{-2} + 5Z^{-3} + Z^{-4} = \frac{1}{Z} + \frac{3}{Z^2} + \frac{5}{Z^3} + \frac{1}{Z^4}$$

$$= \frac{Z^3 + 3Z^2 + 5Z + 1}{Z^4}$$

ROC = All Value of $Z - \{0\}$



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4.3: Find and Plot Z-Transform

① $x(n) = \sin(nT)$, $n = 0, 1, 2 \cdots$

$$X(Z) = \frac{Z \sin T}{Z^2 - 2Z\cos T + 1} \qquad \alpha = T$$

$Z^2 - 2Z\cos T + 1 \qquad a = 1, b = -2\cos T, c = 1$

$$P_{1,2} = \frac{2\cos T \pm \sqrt{4\cos^2 T - 4}}{2} = \frac{2\cos T \pm 2\sqrt{\cos^2 T - 1}}{2}$$

$$= \frac{2\cos T \pm j2\sin T}{2} = \cos T \pm j \sin T$$

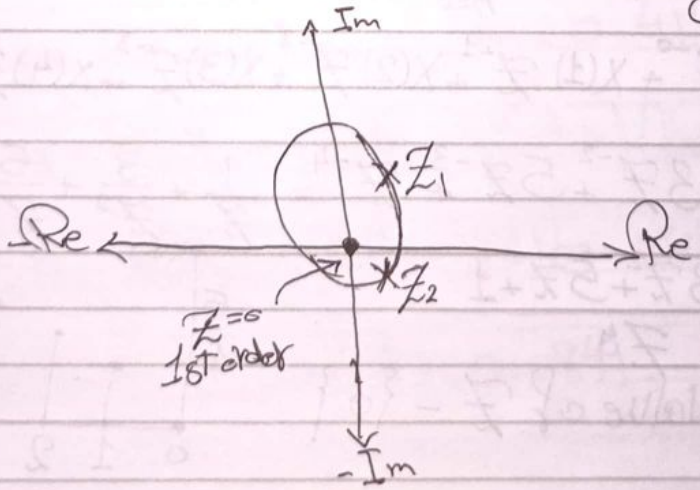
$P_1 = \cos T + j\sin T \qquad P_2 = \cos T - j\sin T$

$$X(Z) = \frac{Z \sin T}{(Z - \cos T - j\sin T)(Z - \cos T + j\sin T)}$$

Poles $Z_1 = \cos T + j\sin T$

$Z_2 = \cos T - j\sin T$

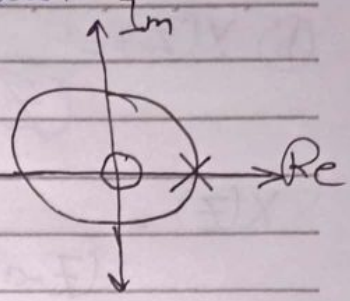
$r = \sqrt{\cos^2 T + \sin^2 T} \qquad \tan\theta = \frac{\sin\theta}{\cos\theta} = \tan T$



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② $x(n) = a^n \quad n > 0$

$X(Z) = \frac{Z}{Z-1}$ have Pole $\rightarrow Z=1$, Zero $\rightarrow Z=0$



4.5: Find using Power Series method First 5 values

ⓐ $X(Z) = \frac{Z-1}{Z^2 - 1.4142Z + 0.4999} \times \frac{Z^{-2}}{Z^{-2}}$

$= \frac{Z^{-1} - Z^{-2}}{1 - 1.4142Z^{-1} + 0.4999Z^{-2}} = \frac{a_0 + a_1Z^{-1} + a_2Z^{-2}}{b_0 + b_1Z^{-1} + b_2Z^{-2}}$

$a_0 \rightarrow 0 \quad b_0 \rightarrow 1$

$a_1 \rightarrow 1 \quad b_1 \rightarrow -1.4142$

$a_2 \rightarrow -1 \quad b_2 \rightarrow 0.4999$

$X(0) = \frac{a_0}{b_0} = Zero$

$X(n) = \left[a_n - \sum_{j=1}^{n} X(n-i)b_i\right] / b_0$

$X(1) = [a_1 - X(0)b_1] / b_0 = 1$

$X(2) = [a_2 - X(1)b_1 - X(0)b_2] / b_0 = 0.4142$

$X(3) = [a_3 - X(2)b_1 - X(1)b_2 - X(0)b_3] / b_0 = 0.0849$

$X(4) = [a_4 - X(3)b_1 - X(2)b_2 - X(1)b_3 - X(0)b_4] / b_0 = -0.0863$

$X(n) = \{0, 1, 0.4142, 0.0849, -0.0863\}$

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(b) $X(Z) = \dfrac{1}{(Z-0.5)(Z+0.9)^3}$

$$X(Z) = \frac{1}{(Z-0.5)(Z+0.9)(Z^2+1.8Z+0.81)}$$

$$= \frac{1}{(Z^2-0.4Z-0.45)(Z^2+1.8Z+0.81)}$$

$$= \frac{1}{Z^4+2.2Z^3+1.08Z^2-0.486Z-0.3645}$$

$$= \frac{Z^{-4}}{1+2.2Z^{-1}+1.08Z^{-2}-0.486Z^{-3}-0.3645Z^{-4}}$$

$a_0 = a_1 = a_2 = a_3 = \text{Zero} \quad a_4 = 1$

$b_0 = 1 \quad b_1 = 2.2 \quad b_2 = 1.8 \quad b_3 = -0.486 \quad b_4 = -0.3645$

$X(0) = \frac{a_0}{b_0} = \text{Zero}$

$X(1) = [a_1 - X(0)b_1] / b_0 = \text{Zero}$

$X(2) = [a_2 - X(1)b_1 - X(0)b_2] / b_0 = \text{Zero}$

$X(3) = [a_3 - X(2)b_1 - X(1)b_2 - X(0)b_3] / b_0 = \text{Zero}$

$X(4) = [a_4 - X(3)b_1 - X(2)b_2 - X(1)b_3 - X(0)b_4] / b_0 = 1$

$$x(n) = \{0, 0, 0, 0, 1\}$$

CamScanner

4.6: Repet Prob 5 using Partial fraction

@ $X(z) = \frac{z-1}{(z-0.707)^2}$

$$\frac{X(z)}{z} = \frac{z-1}{z(z-0.707)^2} = \frac{A}{z} + \frac{B_2}{(z-0.707)^2} + \frac{B_1}{(z-0.707)}$$

$$A = \left.\frac{z-1}{(z-0.707)^2}\right|_{z=0} = -2$$

$$B_2 = \left.\frac{z-1}{z}\right|_{z=0.707} = -0.414$$

$$B_1 = \frac{d}{dz}\left[\frac{z-1}{z}\right] = \left.\frac{z(1) - z + 1}{z^2}\right|_{z=0.707} = \left.\frac{1}{z}\right|_{z=0.707} = 2$$

$$\frac{X(z)}{z} = \frac{-2}{z} + \frac{-0.414}{(z-0.707)^2} + \frac{2}{(z-0.707)}$$

$$X(z) = -2 - \frac{0.414z}{(z-0.707)^2} + \frac{2z}{(z-0.707)}$$

$$x(n) = -2\,\delta(n) - 0.585n(0.707)^n + 2(0.707)^n$$

$x(1) = 1$

$x(2) = 0.414$

$x(3) = 0.0849$

$x(4) = -0.0863$

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4.8. Given diff eq. $y(n) + B_1 y(n-1) + B_2 y(n-2) = A$

Find Z-Transform $Y(z)$ use an iZT To Be obtain a closed form for $y(n)$

$$y(n) + B_1 y(n-1) + B_2 y(n-2) = A$$
$$y(z) + B_1 y(z) z^{-1} + B_2 y(z) z^{-2} = A$$
$$y(z) \cdot [1 + B_1 z^{-1} + B_2 z^{-2}] = A$$

$$y(z) = \frac{A}{1 + B_1 z^{-1} + B_2 z^{-2}}$$

* inverse Z-Transform (Recursively Method)

$$y(z) = \frac{A}{1 + B_1 z^{-1} + B_2 z^{-2}} = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{b_0 + b_1 z^{-1} + b_2 z^{-2}}$$

$a_0 = A \quad a_1 = 0 = a_2$

$b_0 = 1 \quad b_1 = B_1 \quad b_2 = B_2$

$$y(0) = \frac{a_0}{b_0} = A$$

$$y(n) = \left[a_n - \sum_{i=1}^{n} y(n-i) b_i\right] / b_0$$

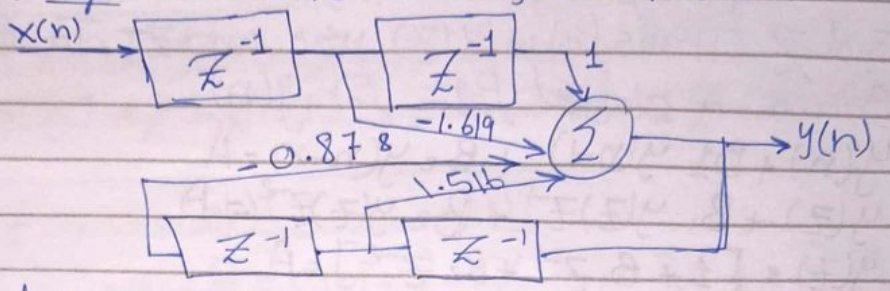
$$y(1) = [a_1 - y(0) b_1] / b_0 = -A B_1$$

$$y(2) = [a_2 - y(1) b_1 - y(0) b_2] / b_0 = A B_1^2 - A B_2$$

$$y(n) = \{A, -AB_1, AB_1^2 - AB_2, \dots\}$$

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4.9: Find diff eq. then drive the System T.F?



$y(n) = -1.619\,x(n-1) + x(n-2) + 1.516\,y(n-1) - 0.878\,y(n-2)$

T.F $\rightarrow H(Z) = \frac{y(Z)}{X(Z)}$

From Diff eq.

$y(Z) = -1.619\,X(Z)Z^{-1} + X(Z)Z^{-2} + 1.516\,Y(Z)Z^{-1} - 0.878\,Y(Z)Z^{-2}$

$= -1.516\,Y(Z)Z^{-1} + 0.878\,Y(Z)Z^{-2} = -1.619\,X(Z)Z^{-1} + X(Z)Z^{-2}$

$Y(Z)\left[1 - 1.516Z^{-1} + 0.878Z^{-2}\right] = X(Z)\left[-1.619Z^{-1} + Z^{-2}\right]$

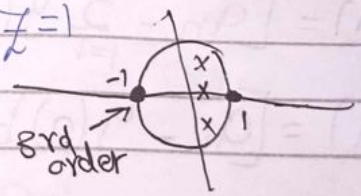
$$\therefore H(Z) = \frac{y(Z)}{X(Z)} = \frac{-1.619Z^{-1} + Z^{-2}}{1 - 1.516Z^{-1} + 0.878Z^{-2}}$$

4.10: Poles $\rightarrow Z = 0.5$, $Z = 0.1 \pm j\,0.02$

Zeros $\rightarrow Z = -1$ (3rd order) , $Z = 1$

ⓐ Sketch Pole-Zero diagram:

↳ System is Stable



ⓑ drive the T.F from Pole-Zero diagram

$$H(Z) = \frac{\text{Zeros}}{\text{Poles}} = \frac{(Z+1)^3(Z-1)}{(Z-0.5)(Z-0.1+0.02j)(Z-0.1-0.02j)}$$

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ⓒ Drive the diff equation:-

- $(z+1)^3(z-1) = (z+1)(z-1)(z^2+2z+1)$
$= (z+1)(z^3+z^2-z-1) = z^4+2z^3-2z-1$ ①*

- $(z-0.5)(z-0.1+0.02j)(z-0.1-0.02j)$
$= (z-0.5)(z^2-0.2z+0.0104) = z^3-0.7z^2+0.1104z-5.2\times10^{-3}$ ②*

$$H(z) = \frac{z^4+2z^3-2z-1}{z^3-0.7z^2+0.1104z-5.2\times10^{-3}}$$

$$\frac{y(z)}{x(z)} = \frac{1+2z^{-1}-2z^{-3}-z^{-4}}{z^{-1}-0.7z^{-2}+0.1104z^{-3}-5.2\times10^{-3}z^{-4}}$$

$$y(z)z^{-1}-0.7y(z)z^{-2}+0.1104y(z)z^{-3}-5.2\times10^{-3}y(z)z^{-4} = x(z)+2x(z)z^{-1}-2x(z)z^{-3}-x(z)z^{-4}$$

$$y(n-1)-0.7y(n-2)+0.1104y(n-3)-5.2\times10^{-3}y(n-4) = x(n)+2x(n-1)-2x(n-3)-x(n-4)$$

ⓓ draw Realization diagram in Signal Flowgraph form.

